Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Build a Human-AI Workforce with Autonomous AI Agents

TGS Ref No: TGS-2024043854

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 2.0

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	12 July 2026	Initial release — 3-day lesson plan for the autonomous-AI-agent labs (Hermes Agent, OpenClaw, Paperclip).	Dr. Alfred Ang
1.1	14 July 2026	Revised to a 2-day plan (Day 1: Topics 1 & 2; Day 2: Topic 3 & final assessment). Hermes expanded to 14 labs, OpenClaw to 12, Paperclip to 8 (34 labs total). Learning outcomes aligned to the Course Proposal / TSC; per-lab reference videos added.	Dr. Alfred Ang
1.2	14 July 2026	Visual redesign: all content slides card-format; YouTube links removed from the deck and Learner Guide (videos remain in the labs); live Hermes screenshots and skill-category grids added; Hermes install lab expanded with the official install methods.	Dr. Alfred Ang
1.3	14 July 2026	Deck restyled in the dark masterclass theme end-to-end; added the Hermes masterclass slides (What is Hermes; The Bet: Compounding Value; platform matrix; Top-10 slash commands; Telegram setup; memory L1-vs-L2 and session-search; what-is-a-tool, built-in tool catalog, tools-in-action, cron scheduling, delegation, kanban orchestration and OpenClaw topic slides); Lab 13 rebuilt as the multi-agent video team (profiles + kanban) with detailed Labs 11-14 guides; fade slide transitions and the live Achievements dashboard screenshot.	Dr. Alfred Ang
1.4	15 July 2026	Security masterclass slides added to the Hermes Security lab (defense-in-depth layers; the gateway's six-check default-deny trust chain; channel lockdown dials; the three approval modes + the Tirith scanner; a live Dangerous-Command approval; YOLO mode; the unrecoverable-command blocklist floor; prompt-injection scanning of context files). Security lab steps expanded with the channel allowlist and approvals.mode.	Dr. Alfred Ang
1.5	15 July 2026	Sixteen concept diagrams imported from the original Mastering OpenClaw masterclass deck into Topic 2 (OpenClaw OS control plane; deployment; hub-and-spoke gateway; tools vs skills; SKILL-on-demand; 6-phase execution loop; exec; HEARTBEAT + autonomic tick; workspace file persistence; context assembly; compaction; defense methods; poisoned-skill anatomy; sessions_spawn; sub-agent routing).	Dr. Alfred Ang
1.6	15 July 2026	Topic 3 (Paperclip) rebuilt as the 'Tertiary AI News Research' company across 6 labs (27-32): install on Windows & Mac; company, CEO & mission; model adaptors; a detailed task backlog whose core task hires the team; the Tavily Search API via Secrets; and hiring the members under the hiring task — with live dashboard screenshots. Natural-language prompt steps added to the Hermes and OpenClaw labs alongside the CLI. Platform logos added to the Three Platforms page. 32 labs total.	Dr. Alfred Ang
1.7	15 July 2026	Sixty-eight content slides migrated from the original 'Mastering OpenClaw' masterclass deck into Topic 2 as framed figures in their matching labs — agent evolution and Agent = Model + Harness, the OpenClaw Wikipedia history and OpenAI acquisition, deployment options and services, model stacks and Mission Control, channels, skills and ClawHub risks, tools (AgentMail, Agent Browser, Firecrawl), system prompt / workspace / context and memory, security lockdown, slash commands, crons, and use cases (NFT certificates, dashboard, cost saving, Google integration).	Dr. Alfred Ang
1.8	15 July 2026	Imported masterclass figures re-blended into the dark deck	Dr. Alfred Ang

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		theme: original title and copyright strips cropped, white canvases converted to the deck background with text inverted to ivory (embedded screenshots untouched), frameless rendering, and duplicated titles reworded.	
1.9	16 July 2026	Deck curated per trainer review: the bulk masterclass figure imports were removed, keeping nine key figures (Wikipedia history, Agent=Model+Harness, SKILL.md, ClawHub, AgentMail, 10-steps lockdown, HEARTBEAT, autonomic tick, workspace persistence) re-exported at high resolution, with the Topic 2 opening story reordered ahead of Key Concepts. Every CLI step now also shows the natural-language prompt alternative ('Or just ask') across the Hermes and OpenClaw labs, in the deck and Learner Guide.	Dr. Alfred Ang
1.10	16 July 2026	PP assessment updated to v4 — evidence screenshots changed to: Task 1 Hermes interacting on Telegram or the multi-agent Kanban board; Task 2 OpenClaw on Telegram or the TUI; Task 3 the Paperclip setup (company, mission, CEO, hired agents) — with Topic 3 lab references refreshed. A 'RAG vs a Standard LLM Assistant' concept slide added to Core Concepts to ground WA Q2. Final PPT/LP/LG/labs/assessment alignment review.	Dr. Alfred Ang
1.11	16 July 2026	Topic 3 company renamed to 'Altera AI Blogs' with the mission re-pointed at publishing AI-related blogs end to end; the core backlog task is now 'Hire the agents to publish the AI-related blogs end to end' and Lab 32's hiring happens explicitly under it; Lab 29 (Adaptor) and Lab 32 descriptions expanded in full; NEW Lab 33 'Routine & Trigger — Publish Daily at 3pm' (33 labs total), verified with a live 'Run now' pipeline test and two live screenshots.	Dr. Alfred Ang
2.0	18 October 2026	Major revision. Deck redesigned in the 'Agentic AI Applications with Claude Code' house design. Topic 1 rebuilt as Foundations: history of AI, AI agents vs agentic AI, the agentic loop, harness engineering, the brain–perception–action framework, short/long-term memory, world models, single → multi-agent → human–agent collaboration paradigms, agent society, the Oct 2026 agent market (OpenClaw, Hermes Agent, Paperclip, OpenAI Dots, Meta Muse, Block Buzz, Manus Cue and more), Singapore's AgentSea (12,000+ agents) and human–AI use cases in science, maths, drug discovery and business. OpenClaw labs removed; 33 feature labs replaced by 10 connected use-case labs for the fictional Kopi Kaki Co. — Day 1 Hermes Agent (research brief, end-to-end social media campaign, human-in-the-loop publishing, multi-agent video studio, campaign command centre) and Day 2 Paperclip (found the company, hire the team, launch project, weekly video routine, govern & improve) — with mock data, brand assets, skills, instruction PDFs and a local Lab Prompts page. Assessment v5 aligned; FG and AP added.	Dr. Alfred Ang

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Course Information

Course Title	Build a Human-AI Workforce with Autonomous AI Agents
WSQ Course Reference	TGS-2024043854
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days · 8 training hours per day (16 hours), including 2 hours of assessment
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks within training time)
Mode	Instructor-led; hands-on use-case labs on Hermes Agent and Paperclip (learner laptop)
Course scenario	Kopi Kaki Co. — a fictional Singapore kopi chain launching the Gula Melaka Cold Brew (#ShiokColdBrew) — all company data is fictional
Slide deck	Human-AI Workforce with Autonomous AI Agents-v2.0.pptx (153 slides)
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Analyse AI (LLM-based) applications across a range of industries to identify their capabilities and limitations.
- LO2: Establish the relationship between AI algorithm/agent design and chatbot/agent efficiency.
- LO3: Evaluate and improve the effectiveness of AI (RAG and agent) applications in product development.

Topics and Labs

Topic	Content / labs	Slides
Topic 1: Foundations of Autonomous AI Agents & the Human–AI Workforce	A brief history of AI; AI agents vs agentic AI; The agentic loop & harness engineering; Brain · perception · action, memory & world models; From single agent to human–AI collaboration; The 2026 agent market, use cases & governance. Activity: Human–AI Workforce Canvas.	20–60
Topic 2: Hermes Agent — From a Single AI Employee to a Multi-Agent Team	Lab 1 · Your first AI employee; Lab 2 · End-to-end social media campaign; Lab 3 · Human-in-the-loop publishing; Lab 4 · Multi-agent video studio; Lab 5 · Campaign command centre.	61–104
Topic 3: Paperclip — Running & Governing a Company of AI Agents	Lab 6 · Found the AI company; Lab 7 · Hire the team; Lab 8 · Launch campaign project; Lab 9 · Weekly video routine; Lab 10 · Govern, measure & improve.	105–145

Learning Units (approved Course Proposal TPG-2024060313)

LU	Learning unit	Delivered through	K & A	Duration
LU1	Foundations of autonomous AI agents and single-agent LLM applications	Topic 1 (Foundations) + Topic 2 Labs 1–2 (single Hermes agent)	K1, K6, A1, A3	5 h
LU2	Agent and multi-agent design for efficiency, with humans in the loop	Topic 2 Labs 3–5 (human-in-the-loop, multi-agent studio, swarm) + Topic 3 Lab 6 (designing a governed agent	K4, K5, A2, A6	5 h

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		company)		
LU3	Evaluate and improve the effectiveness of an AI-agent workforce in product development	Topic 3 Labs 7–10 (team, launch project, routine, governance and improvement)	K2, K3, A4, A5	4 h
Assessment	WA-SAQ (1 h) + PP (1 h)	Day 2, 4:30–6:30 pm	K1–K6, A1–A6	2 h

Total: 16 hours = LU1 5 h + LU2 5 h + LU3 4 h + assessment 2 h, matching the approved Course Proposal.

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.
- Practical Performance (PP) — three tasks, each with ONE screenshot from a lab (Lab 1 Hermes agent, Lab 4 multi-agent studio, Lab 7 Paperclip org chart) plus written analysis, 1 hour, open book.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- Final assessment is conducted on Day 2 from 4:30 pm.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Course Schedule

Day 1 — Foundations (Topic 1) & Hermes Agent — single to multi-agent (Topic 2, Labs 1–5)

Time	Duration	Topic / Activity	LU	Slides
9:30–9:50	20 min	Welcome, housekeeping, trainer introduction, learning outcomes and AM digital attendance	LU1	1–19
9:50–11:05	75 min	**Topic 1 (Part A) — history of AI; AI agents vs agentic AI; the agentic loop; harness engineering; brain–perception–action; short- and long-term memory; RAG; world models**	LU1	20–39
11:05–11:20	15 min	Tea break	LU1	—
11:20–12:10	50 min	**Topic 1 (Part B) — single agent → multi-agent → human–agent collaboration; levels of autonomy; agent society; the 2026 agent market (OpenClaw, Hermes, Paperclip, OpenAI Dots, Meta Muse, Block Buzz, Manus Cue …); Singapore's AgentSea; human–AI use cases in science, maths, drug discovery and business; risks and governance**	LU1	40–58
12:10–12:35	25 min	**Activity — Human–AI Workforce Canvas for Kopi Kaki Co., and start the Hermes Agent install**	LU1	59–60
12:35–13:30	55 min	Topic 2 — Lab 1: Hire Your First AI Employee (Hermes) — customer & market research brief	LU1	61–73
13:30–14:30	60 min	Lunch break	—	—
14:30–15:30	60 min	Lab 2: End-to-End Social Media Campaign with one agent (skills + tools)	LU1	74–80
15:30–16:15	45 min	Lab 3: Human-in-the-Loop Publishing & Community Care (Telegram + cron)	LU2	81–88
16:15–16:30	15 min	Tea break	LU2	—
16:30–17:30	60 min	Lab 4: Multi-Agent Video Studio — research → script →	LU2	89–96

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		production → audit → publish		
17:30–18:15	45 min	Lab 5: Multi-Agent Campaign Command Centre — measure, optimise & report	LU2	97–103
18:15–18:30	15 min	Day 1 recap, Q&A and PM digital attendance	LU2	104

Total training time: 480 minutes (8 hours).

Day 2 — Paperclip — a governed company of agents (Topic 3, Labs 6–10) & Final Assessment

Time	Duration	Topic / Activity	LU	Slides
9:30–9:45	15 min	Day 1 recap and AM digital attendance	LU2	104
9:45–10:15	30 min	**Topic 3 — Paperclip: the company as the agent harness; Board, CEO and org chart; adapters; goals, projects and tasks; routines and heartbeats; budgets, approvals and the audit trail**	LU2	105–109
10:15–11:15	60 min	Lab 6: Found Kopi Kaki Growth Studio — an AI company you govern	LU2	110–116
11:15–11:30	15 min	Tea break	LU2	—
11:30–12:15	45 min	Lab 7: Hire the Marketing Team through the Board approval gate	LU3	117–124
12:15–13:15	60 min	Lunch break	—	—
13:15–14:10	55 min	Lab 8: Run the #ShiokColdBrew launch as a project — review, revise, release	LU3	125–131
14:10–14:55	45 min	Lab 9: Automate the weekly video pipeline with a routine and a Board gate	LU3	132–137
14:55–15:10	15 min	Tea break	LU3	—
15:10–15:55	45 min	Lab 10: Govern, measure & improve your human–AI workforce	LU3	138–144
15:55–16:15	20 min	Course recap, practice exam and Q&A	LU3	146–149
16:15–16:30	15 min	**Briefing for Assessment**	LU3	17
16:30–17:30	60 min	**Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book**	Assessment	150–153
17:30–18:30	60 min	**Practical Performance (PP) — 1 hour, open book; TRAQOM survey and PM / assessment digital attendance**	Assessment	150–153

Total training time: 480 minutes (8 hours).

Lab Reference

Lab	Title	Paradigm	LO	Slides
Lab 1 (Day 1)	Hire Your First AI Employee — Customer & Market Research Brief	Single agent	LO1	67–73
Lab 2 (Day 1)	End-to-End Social Media Campaign with One Agent	Single agent + skills	LO1	74–80
Lab 3 (Day 1)	Human-in-the-Loop Publishing & Community Care (Telegram + Cron)	Human–agent collaboration	LO2	81–88
Lab 4 (Day 1)	Multi-Agent Video Studio — Research → Script → Production → Audit → Publish	Multi-agent (ordered pipeline)	LO2	89–96
Lab 5 (Day 1)	Multi-Agent Campaign Command Centre — Measure, Optimise & Report	Multi-agent (parallel swarm)	LO2	97–103
Lab 6 (Day 2)	Found Kopi Kaki Growth Studio — an AI Company You Govern	Human-governed company of agents	LO2	110–116
Lab 7 (Day 2)	Hire the Marketing Team Through the Board Approval Gate	Multi-agent organisation	LO3	117–124
Lab 8 (Day 2)	Run the #ShiokColdBrew Launch as a Project — Review, Revise, Release	Multi-agent project with Board review	LO3	125–131
Lab 9 (Day	Automate the Weekly Video Pipeline with a Routine	Scheduled multi-agent	LO3	132–137

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2)	and a Board Gate	workflow		
Lab 10 (Day 2)	Govern, Measure & Improve Your Human–AI Workforce	Workforce governance	LO3	138–144

Lab materials: labs/labNN-*/ (README, LAB-NN-Instructions.pdf, COPY-PROMPTS.txt, data/) and the Lab Prompts page prompts/index.html in the course repository.

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